Ashan Odithya

Software Engineer | Node.js · React · TypeScript · AWS

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Summary

Software Engineer (Full Stack) with production experience in full-stack development of web applications using Node.js, Express.js, React and TypeScript on AWS services (Lambda, API Gateway, DynamoDB, S3, Cognito) at WealthOS, a JPMorgan company. Develops and consumes RESTful APIs, builds serverless backend services within a microservices architecture, and secures APIs with JWT and OAuth 2.0 authentication. Uses Git version control with code reviews, and delivers through CI/CD with QA, DevOps and senior engineers in Agile development across the full SDLC. Known for clear communication and strong analytical and problem-solving skills.

Professional experience

WealthOS (a JPMorgan Company) *Nov 2025 - Present*

Software Engineer Intern *Colombo, Sri Lanka*

Product Development — Wellness Allowance Platform

- Built and deployed a serverless employee wellness benefits platform for submitting, reviewing, approving, and tracking healthcare and wellness expense claims.
- Developed full-stack features in React and Node.js (TypeScript) on AWS Lambda for claim submission, validation, balance tracking and multi-stage HR/Finance approvals, serving 75 employees and processing ∼300 claims per month.
- Designed and built 18 RESTful API endpoints as serverless services on AWS Lambda and API Gateway with DynamoDB, configuring API Gateway integrations and documenting them in OpenAPI/Swagger.
- Implemented API authentication and authorization with Amazon Cognito (OAuth 2.0, JWT) and role-based access control, and secured document uploads using S3 presigned URLs.
- Automated multi-environment deployments with CI/CD and Terraform, and supported production through debugging, root cause analysis, testing and fixes.

Production Engineering & Agile Delivery

- Investigated and resolved production defects through root cause analysis, and delivered fixes and enhancements that improved stability, maintainability and performance without regressions.
- Collaborated with senior engineers, QA, DevOps and Product Owners to troubleshoot issues, validate fixes and coordinate deployments and production releases.
- Participated in code reviews through Git/Bitbucket pull-request workflows, following team coding standards and incorporating peer feedback.
- *Tech Stack:* TypeScript, Node.js, React, AWS Lambda, API Gateway, DynamoDB, S3, Cognito, REST APIs, Terraform

Notable projects & research

Lanka Card Deals: Credit Card Offers Aggregator *Live*

- Built a full-stack aggregator that scrapes, normalizes and serves ∼1,080 live credit card offers from 3 Sri Lankan banks, with daily automated updates, search, filters, sorting and a cross-bank comparison page.

- Developed TypeScript scrapers for a JSON API and server-rendered HTML; reverse-engineered a bank's frontend bundle to find its full-list endpoint, raising coverage from a random 20 to the full ∼900-offer list.
- Handled AWS WAF rate limiting and CI IP blocking with throttled requests, a 403 circuit breaker and split scheduling (GitHub Actions and a secret-protected Vercel Cron route), plus a guard that stops a broken scrape from expiring live offers.
- Validated every offer with Zod before MongoDB bulk upserts, and built a discount classifier that separates real discounts from payment-plan fees, covered by 31 unit tests on real bank data in a CI pipeline (type-check, tests, build).
- Built the Next.js App Router frontend with Server Components, URL-based filters and pagination, and a JSON REST API, deployed on Vercel and MongoDB Atlas free tiers.
- *Tech Stack:* TypeScript, Next.js, React, Node.js, MongoDB Atlas, Mongoose, Zod, Tailwind CSS, REST APIs, GitHub Actions, Vercel, Web Scraping

POP Cosmetics: Full-Stack E-Commerce Platform *Link*

- Built a full-stack MERN e-commerce app with product search, filtering, cart, wishlist, checkout, order history, and reviews.
- Designed REST APIs with JWT authentication, role-based access control (customer/admin), and rate limiting on sensitive routes.
- Prevented overselling with atomic MongoDB stock updates; implemented order status workflows and an admin dashboard.
- Integrated Google Gemini for an AI shopping assistant, using structured JSON output and server-side validation of every recommendation.
- Built a responsive React and Tailwind UI with dark mode and Context-based state management.
- *Tech Stack:* MongoDB, Mongoose, Express.js, React, Node.js, JavaScript, JWT, REST APIs, Tailwind CSS, Vite, Git

PopcornPicks: Movie Explorer Web App *Live*

- Built a movie discovery app on the TMDb API with login, infinite-scroll search, trailers and filters.
- Responsive Material UI interface with dark mode, favorites saved locally, deployed on Vercel.
- *Tech Stack:* React, TypeScript, Material UI, React Router, Axios, Context API, Vite

Kaalagune: Secure Weather Dashboard *Live*

- Built a weather app on the OpenWeather API with live conditions, hourly and 5-day forecasts, air quality and city search.
- Secured with Auth0 (OAuth 2.0 / OIDC) login and multi-factor authentication, plus a guest mode; cached API data to cut requests.
- Responsive React and Tailwind UI with a custom SVG chart and weather-based themes, deployed on Vercel.
- *Tech Stack:* React, JavaScript, Auth0, OpenWeather API, Axios, Tailwind CSS, Motion, Vercel

Forge: Full-Stack Issue Tracking System *Link*

- Built a Jira-style issue tracker with a Spring Boot REST API and React frontend to manage customer issues through a support workflow.
- Designed a layered backend with Spring Data JPA and Hibernate, offering full CRUD and an audit trail of every status change.

- Built a Kanban board with search and rule-based status transitions, plus a Chart.js analytics dashboard.
- Created reusable shadcn/ui-style components in Tailwind CSS, with route guards and error handling.
- Set up MySQL through environment variables, plus an embedded H2 profile for running locally with no database install.
- *Tech Stack:* Java, Spring Boot, Spring Data JPA, Hibernate, MySQL, H2, Maven, React, JavaScript, Tailwind CSS, Chart.js, Axios, REST APIs, Vite

QoS-Aware Distributed Query Processing System *Research*

- Developed a distributed SQL query optimization framework using Apache Calcite, PostgreSQL, and LinUCB, enabling join-based query decomposition and QoS-aware strategy selection.
- Implemented adaptive cut-point selection using data locality, transfer cost, planner statistics, and execution feedback.
- Evaluated using TPC-H, demonstrating improved execution performance over static QoS-based optimization.

Education

BSc. Engineering (Hons.) in Computer Engineering *University of Jaffna*

Relevant coursework: Software Engineering, Database Management Systems, Data Structures & Algorithms, Operating Systems, Computer Networks & Security, Artificial Intelligence, Machine Learning, Deep Learning

Technical skills

Languages	TypeScript, JavaScript, Java, Python, C++
Backend & APIs	Node.js, Express.js, RESTful APIs, API integration, Microservices architecture, Serverless, Middleware, JWT, OAuth 2.0, OpenAPI/Swagger
Frontend	React.js, Redux Toolkit, Tailwind CSS, Material UI, Ant Design, HTML5, CSS3
AWS Services	AWS Lambda, Amazon API Gateway, S3, DynamoDB, Cognito, CloudWatch, VPC, Route 53
Databases	MongoDB, MySQL, PostgreSQL, DynamoDB
Testing	Playwright, unit testing, integration testing, automated testing
Version Control	Git, GitHub, Bitbucket, pull-request code reviews
DevOps & Process	Docker, Terraform, CI/CD, GitHub Actions, Jira, Agile/Scrum, SDLC

Awards & activities

UOJ Coders 4.0	Finalist — Sri Lanka Rank 12 (Team Code_Mazer)
MoraXtreme 9.0	Sri Lanka Rank 82 (Team ViperBytes)

Referees

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